

Forecasting daily store demand

Wesley Gates, September 2026. Summary of a longer report.

A daily, store-level demand forecast, tested on a full year at ten stores against the forecasts a planner would use without it. Public data (the M5 dataset of Walmart daily sales, with a calendar of holidays and SNAP benefit days).

84% of store-weeks better than ordering from last week's number, by a median of 28%, on a fast, regular mover.

10 of 10 stores better than ARIMA, the strongest statistical model, over the full year.

23% less weekly order error than last week's number, 36 units a week per store down to 27.6.

Five years of daily history, ten stores, 358 forecast origins per store, 3,580 scored store-weeks per method, ten methods (two benchmarks, four statistical models and four machine learning variants), every model refit at every origin.

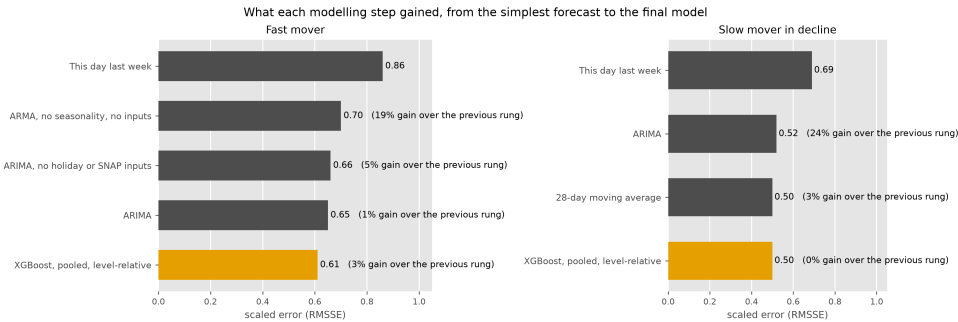


Figure 1. What each modelling step gained, from the simplest forecast (top) to the final model (bottom, gold). The label on each bar is the median gain over the step before it. The final model is a machine learning model, gradient-boosted trees (XGBoost) trained across all ten stores on a level-relative target. Each step is measured against the one before it, so the final model's gain is over a strong statistical model.

Table 1. The fast mover, every day of the test year as a forecast origin, for ten stores. Each rung is scored against the previous rung and against the first, store-week by store-week. RMSE is the scaled error the M5 competition used, and 1.0 is last week's number on the training history. Weekly error is the average miss on a week's total order, in units. For the final model that is about 9% of a typical week's sales, and 11% for last week's number.

rung	RMSSE	weekly error, units	vs the previous rung			vs seasonal naive	
			weeks won	median gain	stores won of 10	weeks won	median gain
This day last week (seasonal naive)	0.86	36.0					
ARMA, no seasonality, no inputs	0.70	33.5	77%	19%	10	77%	19%
ARIMA, no holiday or SNAP inputs	0.66	32.7	65%	5%	10	82%	24%
ARIMA	0.65	31.5	58%	1%	9	83%	25%
XGBoost, pooled, level-relative (final)	0.61	27.6	58%	3%	10	84%	28%

On a slow mover in decline (0.6 to 7 units a day) a 28-day moving average is the best forecast and the final XGBoost model ties it, so items are routed by demand class before anything is fitted, the machine learning model for steady daily movers and the simple average for the rest. The machine learning model still beats last week's number in 82% of store-weeks on this item and ARIMA at all ten stores.

Table 2. The slow mover, the same methods scored against last week's number. The full ladder is in the main report.

method	RMSSE	weekly error, units	weeks won vs seasonal naive	median gain vs seasonal naive
This day last week (seasonal naive)	0.69	5.9		
ARIMA	0.52	5.8	77%	24%
28-day moving average	0.50	5.1	82%	26%
XGBoost, pooled, level-relative (final)	0.50	5.2	82%	26%

What the system does

- Forecasts a seven-day order window from every day of the year, refitting at each origin, the way an ordering system might use them.
- Holiday and SNAP effects measured from the data. Seven of thirty calendar events move this item's sales, and the model sees the days until and since each.
- Bias reported beside error throughout. A forecast that runs systematically high is shrink every week and one that runs low is a stockout.
- Every number reproducible to the byte. Each change to a model is scored against the previous version on the same 3,580 store-weeks, so noise shows up as a coin-flip win rate.
- A full-year evaluation of all methods on one item in about fifteen minutes, and a validator that has to pass before any number is quoted.

The model's inputs are recent sales (the last three days and the same weekday over the last four weeks), the weekly pattern, the level and its spread over the last week to two months, the calendar and how many days ahead the forecast is, the days until and since the seven holidays that matter, SNAP benefit days, and, in the pooled model, the store.

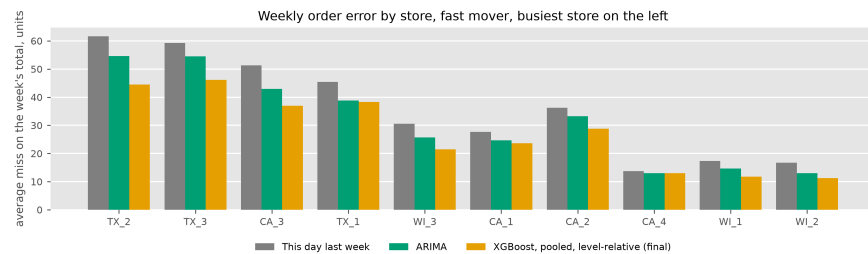


Figure 2. The average miss on a week's total order, in units, by store, busiest on the left. The final model misses by less than last week's number at every store and by less than ARIMA at nine of ten, level at the tenth. The busiest store benefits the most, 17 fewer units a week.

Next steps

- Turn the forecast into an order quantity at a chosen service level, which sets the trade-off between shrink and stockouts once, and score the orders on how often they would have run short.
- Extend the routing rule to intermittent items with Croston's method and its TSB variant, and to new items with no sales history (zero-shot).
- Add the yearly pattern with Fourier terms or an STL decomposition, the approach recommended for daily data with a yearly cycle (Hyndman et al., 2026, §13.1).
- Scale from two items and ten stores to a department of 800 items, with the run records in a shared SQL database.

Full report and code: github.com/wesGates/demand-forecasting (the first study as reported is tag first-study there).

Hyndman, R. J., Athanasopoulos, G., Garza, A., Challu, C., Mergenthaler, M., and Olivares, K. G. (2026). *Forecasting: Principles and Practice, the Pythonic Way*. OTexts.

Makridakis, S., Spiliotis, E., and Assimakopoulos, V. (2022). M5 accuracy competition. *International Journal of Forecasting*, 38(4), 1346–1364.